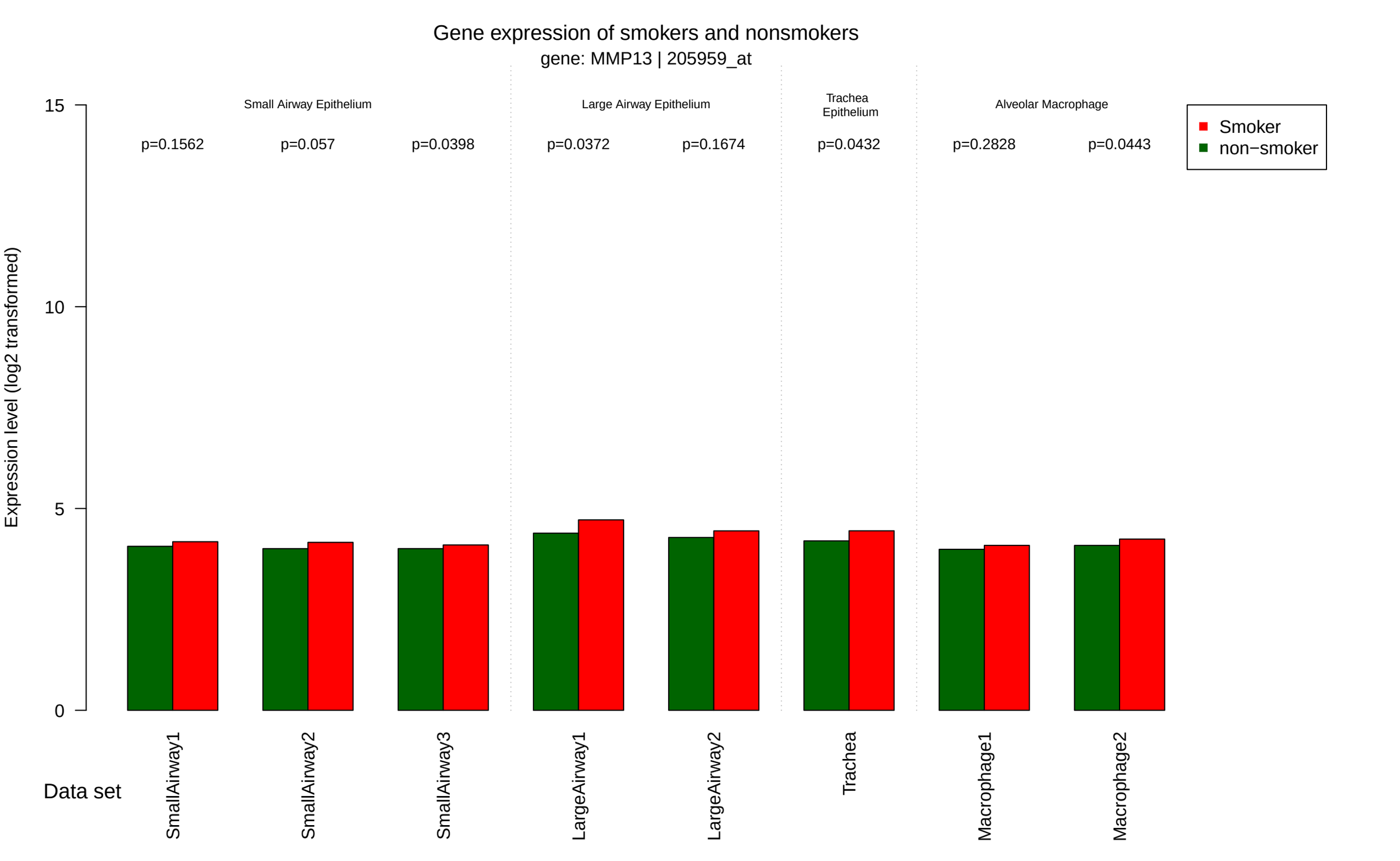
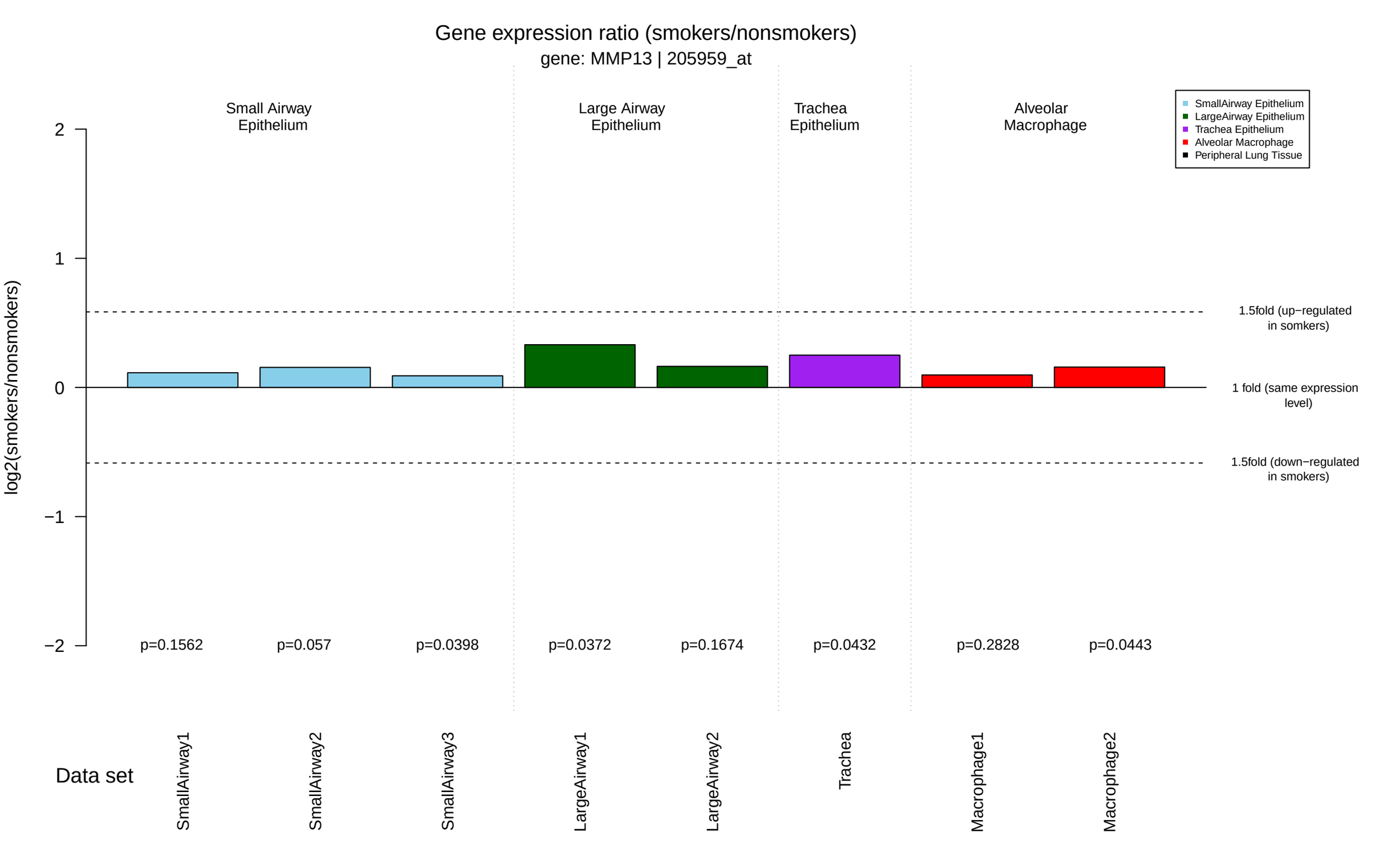
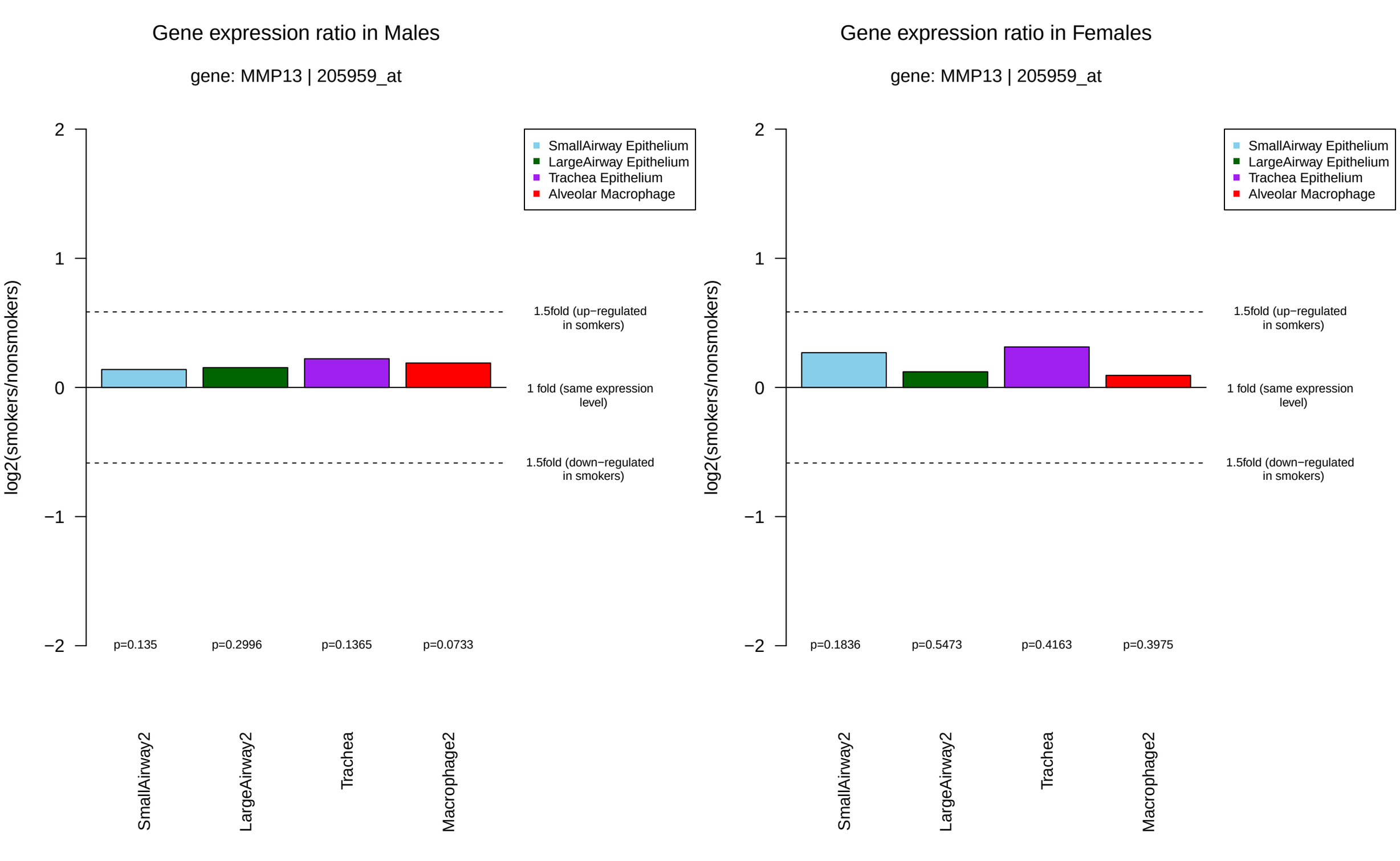
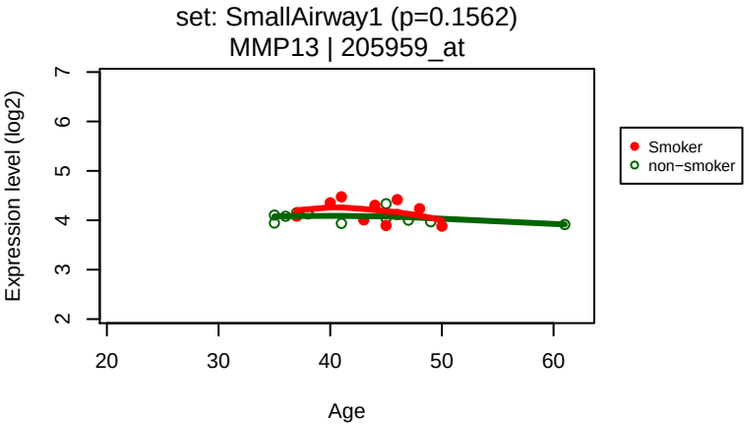
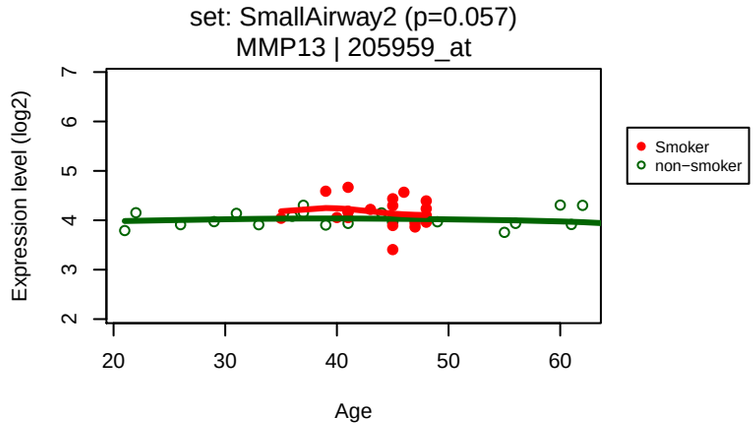
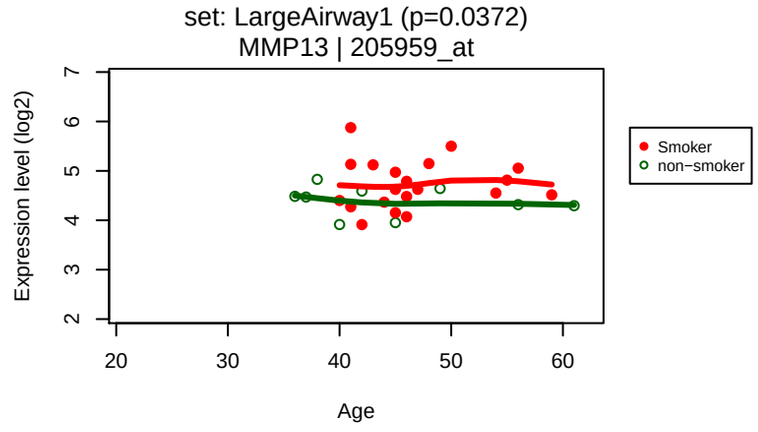

Gene expression of smokers and nonsmokers
gene: MMP13 | 205959_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
p=0.1562
p=0.057
p=0.0398
p=0.0372
p=0.1674
p=0.0432
p=0.2828
p=0.0443
Smoker
non−smoker
Expression level (log2 transformed)
15
10
5
0
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

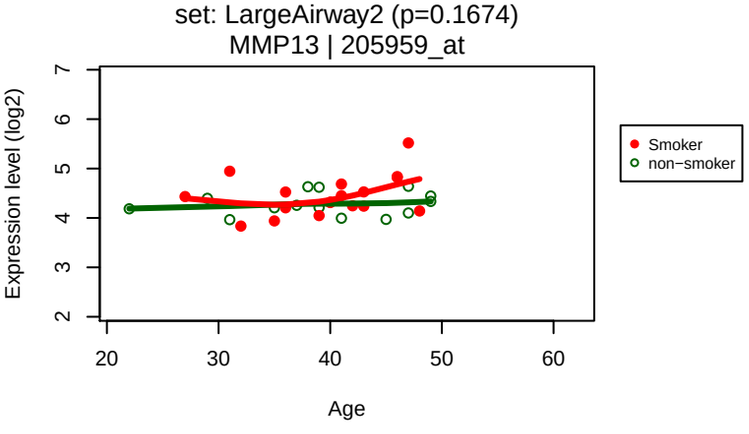

Gene expression ratio (smokers/nonsmokers)
gene: MMP13 | 205959_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Peripheral Lung Tissue
log2(smokers/nonsmokers)
2
1
0
−1
−2
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.1562
p=0.057
p=0.0398
p=0.0372
p=0.1674
p=0.0432
p=0.2828
p=0.0443
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

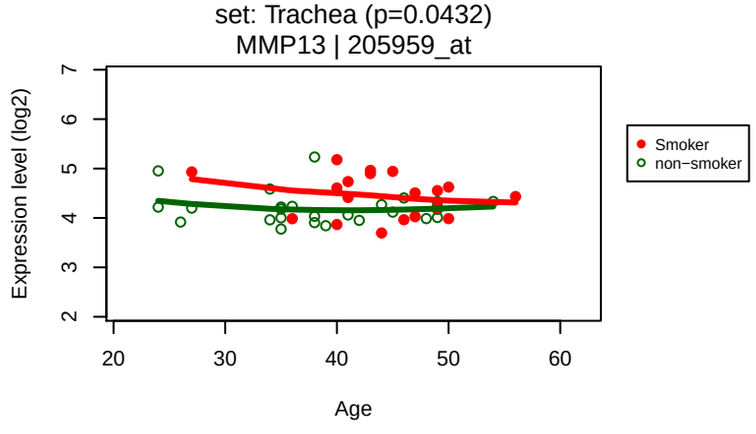

Gene expression ratio in Males
gene: MMP13 | 205959_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.135
p=0.2996
p=0.1365
p=0.0733
SmallAirway2
LargeAirway2
Trachea
Macrophage2
Gene expression ratio in Females
gene: MMP13 | 205959_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.1836
p=0.5473
p=0.4163
p=0.3975
SmallAirway2
LargeAirway2
Trachea
Macrophage2

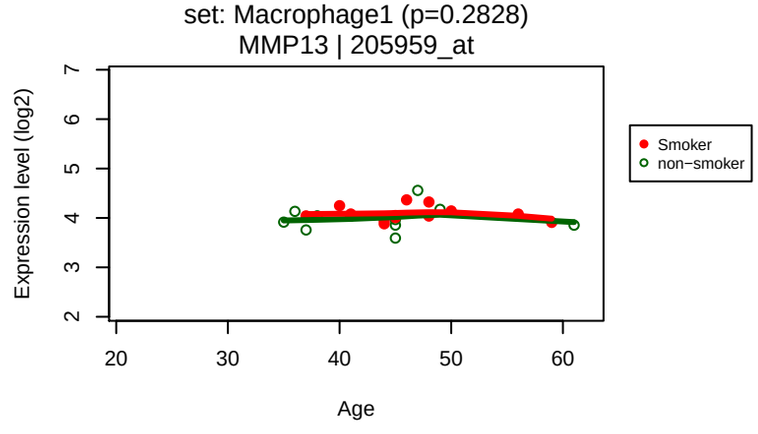

set: SmallAirway1 (p=0.1562)
MMP13 | 205959_at
Expression level (log2)
Age
Smoker
non−smoker

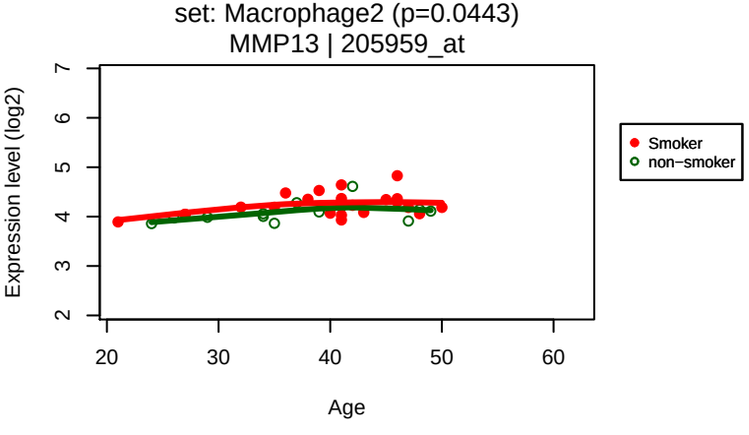

set: SmallAirway2 (p=0.057)
MMP13 | 205959_at
Expression level (log2)
Age
Smoker
non−smoker

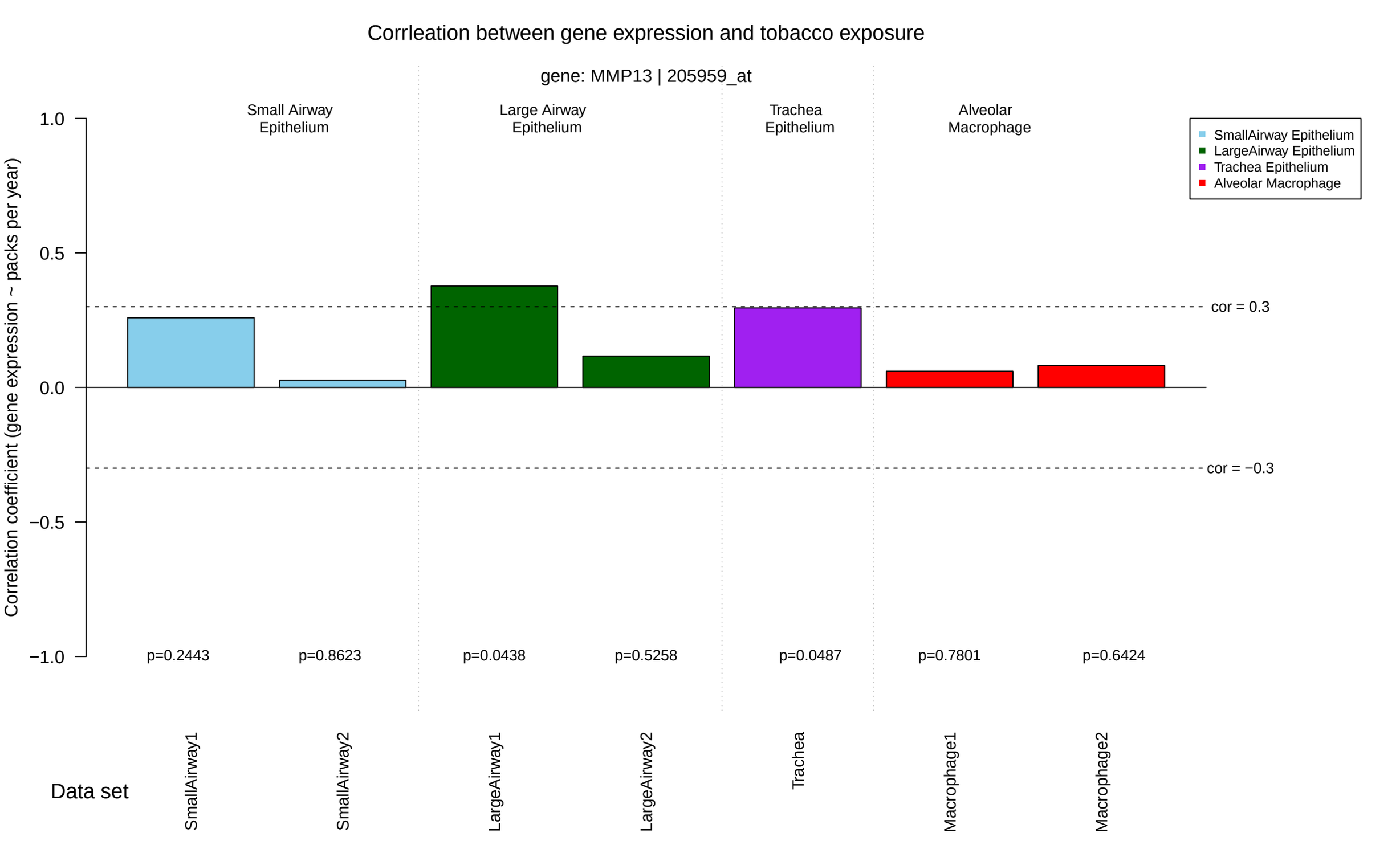

set: LargeAirway1 (p=0.0372)
MMP13 | 205959_at
Expression level (log2)
Age
Smoker
non−smoker


set: LargeAirway2 (p=0.1674)
MMP13 | 205959_at
Expression level (log2)
Age
Smoker
non−smoker


set: Trachea (p=0.0432)
MMP13 | 205959_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage1 (p=0.2828)
MMP13 | 205959_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage2 (p=0.0443)
MMP13 | 205959_at
Expression level (log2)
Age
Smoker
non−smoker


Corrleation between gene expression and tobacco exposure
gene: MMP13 | 205959_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Correlation coefficient (gene expression ~ packs per year)
1.0
0.5
0.0
−0.5
−1.0
cor = 0.3
cor = −0.3
p=0.2443
p=0.8623
p=0.0438
p=0.5258
p=0.0487
p=0.7801
p=0.6424
Data set
SmallAirway1
SmallAirway2
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2